

Task 5: Business Insight & Recommendation Project



1. Introduction

The purpose of this task is to generate meaningful business insights from cleaned e-commerce dataset. This task focuses on using the results from data analysis and visualization to make useful business recommendations.

The dataset contains customer order information, including products, quantities, prices, payment methods, order statuses, coupon codes, referral sources, and total prices.

2. Project Objective

Area	Purpose
Product Performance	Identify high-demand and high-revenue products
Revenue Analysis	Understand which products generate the most money
Order Status	Identify possible issues with cancelled or returned orders
Payment Methods	Understand customer payment preferences
Referral Sources	Identify strong marketing channels
Coupon Usage	Understand customer response to discounts

Table 1: Objectives Table

3. Key Business Questions

This project answers the following business questions:

1. Which products are ordered the most?
2. Which products generate the highest revenue?
3. Which order statuses are most common?
4. Which payment methods are most used by customers?
5. Which referral sources bring the most orders?
6. Which coupon codes are used the most?
7. What recommendations can be made based on these insights?

4. Product Performance Insight

The analysis showed that Printer had the highest number of orders, followed closely by Tablet and Chair.

Product	Number of Orders
Printer	181
Tablet	179
Chair	178
Laptop	173
Desk	170
Monitor	163
Phone	156

Table 2: Number of Orders for Each Product

This shows that Printer, Tablet, and Chair are the most frequently ordered products. These products have strong customer demand.

5. Revenue Insight

The total revenue generated by each product was analysed.

Product	Total Revenue
Chair	195,620.11
Printer	195,612.61
Laptop	192,126.56
Tablet	186,568.95
Monitor	175,651.41
Desk	167,459.93
Phone	151,722.39

Table 3: Total Revenue Generated by Each Product

Chair generated the highest total revenue, followed very closely by Printer. Laptop also generated strong revenue.

This shows that Chair, Printer, and Laptop are important products for the business because they contribute the most to total sales revenue.

6. Order Status Insight

The order status distribution was reviewed to understand how customer orders are being completed.

Order Status	Percentage
Cancelled	21%
Returned	20%
Pending	20%
Shipped	20%
Delivered	19%

Table 4: Order Status Distribution

Cancelled and Returned orders make up a large portion of the dataset. This may be a business concern because these orders may reduce successful sales and customer satisfaction.

The business should investigate why many orders are being cancelled or returned.

7. Payment Method Insight

The payment method distribution shows that customers use different payment options.

Payment Method	Percentage
Online	21%
Cash	21%
Credit Card	20%
Debit Card	19%
Gift Card	19%

Table 5: Payment Method Distribution

Online and Cash payments are the most used methods. However, the differences between payment methods are small, meaning customers are spread across several payment options.

This suggests that the business should continue supporting multiple payment methods.

8. Referral Source Insight

The referral source analysis shows where customers came from before placing orders.

Referral Source	Number of Orders
Instagram	259
Email	250
Google	241
Facebook	228
Referral	222

Table 6: Referral Sources Distribution

Instagram brought the highest number of orders, followed by Email and Google.

This suggests that Instagram and Email are strong marketing channels for the business.

9. Coupon Usage Insight

The coupon code distribution was analysed to understand customer response to discount offers.

Coupon Code	Percentage
FREESHIP	26%
No Coupon	26%
WINTER15	24%
SAVE10	24%

Table 7: Coupon Code Usage Distribution

FREESHIP was one of the most used coupon codes. This suggests that customers may respond well to free shipping offers.

At the same time, many customers placed orders without using a coupon. This shows that not all purchases depend on discounts.

10. Business Recommendations

Based on the insights, the following recommendations can be made:

Recommendation	Reason
Focus on Chair, Printer, and Laptop	These products generated the highest revenue
Promote Printer, Tablet, and Chair	These products had the highest number of orders
Investigate cancelled and returned orders	Cancelled and returned orders are common and may affect business performance
Strengthen Instagram and Email marketing	These were the strongest referral sources
Continue using free shipping offers	FREESHIP was one of the most used coupon codes
Keep multiple payment options available	Customers use a variety of payment methods

Table 8: Recommendations Table

11. Conclusion

In this task, the cleaned e-commerce dataset was used to generate business insights and recommendations.

The analysis showed that Printer had the highest number of orders, while Chair generated the highest total revenue. Instagram was the strongest referral source, and FREESHIP was one of the most used coupon codes.

The analysis also showed that cancelled and returned orders were common, which may be a concern for the business.

Overall, the business should focus on high-performing products, improve order completion, continue using strong marketing channels, and offer useful promotions such as free shipping.